EX-MA CABLE EXCAVATOR

WOOD CUT LIST

Every plywood piece for the sandbox, the pedestal and the control box, with sheet layouts, cut order and drilling — for **A**, the design in the assembly manual, and **B**, a minimal version with a compact control box and no 2in PVC conduit.

HOW MUCH WOOD

Group	A (sq in)	B (sq in)
Sandbox (5 pieces)	1430	1430
Pedestal	294	280
Control box	759	485
Spacer frames	0	99
Total	**2482**	**2294**
Sheets (4 × 8 ft)	1	1

WHAT TO BUY

- **One 4 × 8 ft sheet of 1/2″ plywood** (sanded 15/32″ or 12 mm Baltic birch) covers either version; ask the store to make the first rip (R1) if the sheet will not fit your car.
- Project panels work too: the sandbox floor is exactly a **2 × 2 ft** panel (24″ — the CAD size is 1/64″ more, irrelevant); the rest of either version fits on **two 2 × 4 ft** panels plus the offcut of a third.
- Sizes assume 12 mm (15/32″) stock. With true 1/2″ the boxes come out up to 1/16″ bigger outside — no change needed.

BEFORE YOU CUT

- All sizes in **inches**, to the nearest 1/16″. The small grey mm figures are the exact CAD sizes, for metric tapes.
- Layouts allow a 1/8″ saw kerf between pieces. Holes are given as standard drill-bit sizes.
- IDs **W01–W21** are the same as in the assembly manual; W22/W23 exist only in version B.
- Write the ID on every piece in pencil as soon as it is cut, and mark its **bottom** edge — hole positions are measured from it.
- Drill the box front (W02) and the sandbox wall –u (W18) **clamped together** so their holes line up.

VERSION B IN ONE PARAGRAPH

Control box 8-7/16″ × 14-1/4″ × 6-1/8″ (the 2026-09-28 compact spec). Pedestal 1/2″ shorter on its +u side and pushed 2-1/4″ toward the box; a two-layer plywood spacer (W22, W23) closes the gap to the sandbox wall instead of the PVC conduit. Page 6 explains it.

VERSION A

AS IN THE ASSEMBLY MANUAL

The modelled design: control box 9-13/16″ × 15-15/16″ × 9-1/2″, pedestal 7-7/8″ × 7-1/16″, sandbox 24″ × 24″ × 9-1/2″, joined by a 5-9/16″ length of 2in PVC conduit. Sizes and holes from `build-box.py` (same as manual Appendix B).

2482 sq in
plywood in the pieces (17.2 sq ft)

1 × 4×8 ft
sheet to buy (54 % used)

21 pieces
Control box 759 · Pedestal 294 · Sandbox 1430 sq in

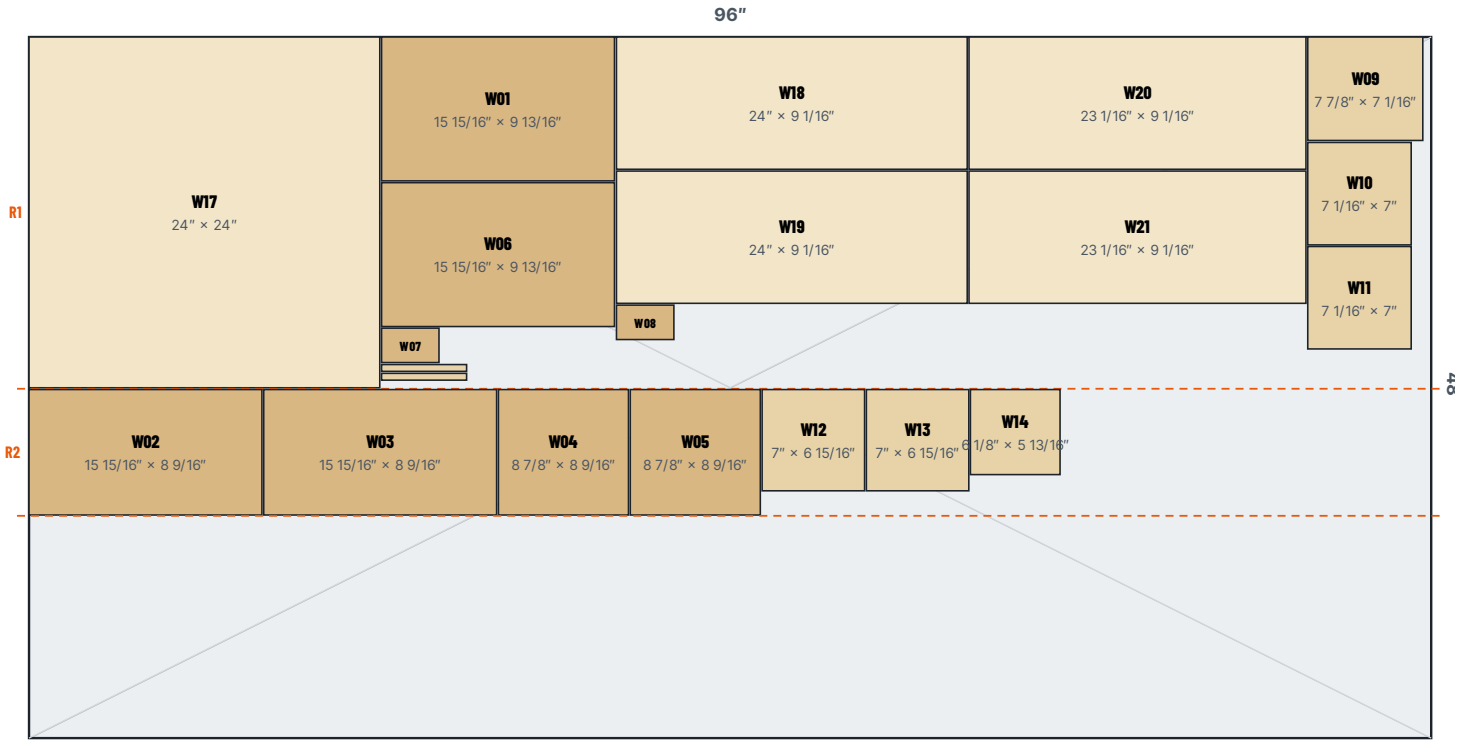
	ID	Piece	Cut size (in)	mm	Holes	sq in
☐	W01	box floor	**15 15/16″ × 9 13/16″**	405 × 250	1	157
☐	W02	box front	**15 15/16″ × 8 9/16″**	405 × 218	7	137
☐	W03	box rear	**15 15/16″ × 8 9/16″**	405 × 218	—	137
☐	W04	box right side	**8 7/8″ × 8 9/16″**	226 × 218	—	76
☐	W05	box left side	**8 7/8″ × 8 9/16″**	226 × 218	—	76
☐	W06	box top (removable)	**15 15/16″ × 9 13/16″**	405 × 250	4	157
☐	W07	axle bearing block left	**3 15/16″ × 2 3/8″**	100 × 60	1	9
☐	W08	axle bearing block right	**3 15/16″ × 2 3/8″**	100 × 60	1	9
☐	W09	pedestal top	**7 7/8″ × 7 1/16″**	200 × 180	5	56
☐	W10	pedestal wall -u (conduit)	**7 1/16″ × 7″**	180 × 178	5	50
☐	W11	pedestal wall +u (removable access)	**7 1/16″ × 7″**	180 × 178	—	50
☐	W12	pedestal wall -v	**7″ × 6 15/16″**	178 × 176	—	49
☐	W13	pedestal wall +v	**7″ × 6 15/16″**	178 × 176	—	49
☐	W14	pedestal shelf	**6 1/8″ × 5 13/16″**	156 × 148	1	36
☐	W15	pedestal shelf cleat -v	**5 13/16″ × 1/2″**	148 × 12	—	3
☐	W16	pedestal shelf cleat +v	**5 13/16″ × 1/2″**	148 × 12	—	3
☐	W17	sandbox floor	**24″ × 24″**	610 × 610	—	577
☐	W18	sandbox wall -u (box side)	**24″ × 9 1/16″**	610 × 230	7	217
☐	W19	sandbox wall +u	**24″ × 9 1/16″**	610 × 230	—	217
☐	W20	sandbox wall -v	**23 1/16″ × 9 1/16″**	586 × 230	—	209
☐	W21	sandbox wall +v	**23 1/16″ × 9 1/16″**	586 × 230	—	209

VERSION A

SHEET LAYOUT AND CUT ORDER

Rip the strips first (orange dashed lines, R1, R2 …), then crosscut each strip. Every piece is drawn to scale with 1/8″ kerf between pieces; grey is offcut. Mark each piece with its ID as soon as it is cut.

SHEET 1 OF 1 — 4 × 8 FT, 1/2″ PLYWOOD



Thin strips not labelled on the drawing: W15, W16 — see the cut order.

R1: rip a strip **24″** wide; crosscut: 24″ → **W17**; 15 15/16″ → **W01** + **W06** + **W07** + **W15** + **W16** (rip that piece into 9 13/16″ / 9 13/16″ / 2 3/8″ / 1/2″ / 1/2″); 24″ → **W18** + **W19** + **W08** (rip that piece into 9 1/16″ / 9 1/16″ / 2 3/8″); 23 1/16″ → **W20** + **W21** (rip that piece into 9 1/16″ / 9 1/16″); 7 7/8″ → **W09** + **W10** + **W11** (rip that piece into 7 1/16″ / 7″ / 7″).

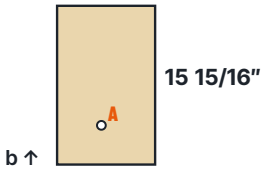
R2: rip a strip **8 9/16″** wide; crosscut: 15 15/16″ → **W02**; 15 15/16″ → **W03**; 8 7/8″ → **W04**; 8 7/8″ → **W05**; 7″ → **W12**; 7″ → **W13**; 6 1/8″ → **W14**.

VERSION A

DRILLING

Measure **a** along the bottom edge and **b** up the side, from the lower-left corner as drawn. Holes: the bold size is the drill bit to use. Clamp a scrap under the panel when drilling the big holes (2-7/16″ hole saw).

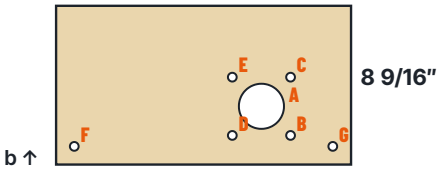
W01 box floor



a from the −u edge, b from the −v (right) edge — seen from above

	Hole / bit	a / b (in)	mm	For
A	**11/32″** Ø8.5 mm	**4 7/16″ / 3 15/16″**	113 / 100	slew spool axle (nut + washer in a Ø22 × 7 counterbore from below)

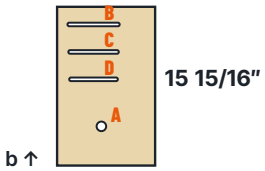
W02 box front



a from the −v (right) edge, b from the bottom edge — seen from the +u side

	Hole / bit	a / b (in)	mm	For
A	**2-7/16″ hole saw** Ø62.0 mm	**11 1/8″ / 3 1/8″**	282 / 80	2in conduit
B	**3/16″** Ø4.6 mm	**12 11/16″ / 1 9/16″**	322 / 40	fitting bolt (M3 × 35 through the sandbox wall)
C	**3/16″** Ø4.6 mm	**12 11/16″ / 4 3/4″**	322 / 120	fitting bolt (M3 × 35 through the sandbox wall)
D	**3/16″** Ø4.6 mm	**9 1/2″ / 1 9/16″**	242 / 40	fitting bolt (M3 × 35 through the sandbox wall)
E	**3/16″** Ø4.6 mm	**9 1/2″ / 4 3/4″**	242 / 120	fitting bolt (M3 × 35 through the sandbox wall)
F	**3/16″** Ø4.6 mm	**1″ / 1″**	25 / 25	box-to-sandbox bolt (M3 × 30)
G	**3/16″** Ø4.6 mm	**14 15/16″ / 1″**	380 / 25	box-to-sandbox bolt (M3 × 30)

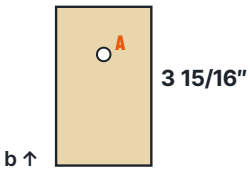
W06 box top (removable)



a from the −u edge, b from the −v (right) edge — seen from above

	Hole / bit	a / b (in)	mm	For
A	**1″ spade / Forstner** Ø24.4 mm	**4 7/16″ / 3 15/16″**	113 / 100	spool tube bearing
B	**slot 5 1/4″**: 23/64″ holes at a 1 1/4″ & 6 1/8″, saw between 133.0 mm	**3 11/16″ / 14 3/16″**	93 / 360	boom lever slot = hard stop (42 deg swing)
C	**slot 5 1/16″**: 23/64″ holes at a 1 5/16″ & 6″, saw between 128.7 mm	**3 11/16″ / 11 7/16″**	93 / 290	stick lever slot = hard stop (41 deg swing)
D	**slot 4 7/8″**: 23/64″ holes at a 1 3/8″ & 5 15/16″, saw between 123.9 mm	**3 11/16″ / 8 11/16″**	93 / 220	bucket lever slot = hard stop (40 deg swing)

W07 + W08 axle bearing blocks (×2, identical)



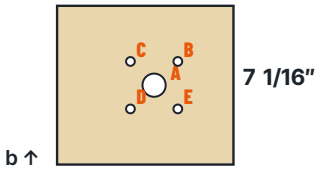
a from the −u edge, b from the bottom edge — seen from the −v side

	Hole / bit	a / b (in)	mm	For
A	**11/32″** Ø8.5 mm	**1 3/16″ / 2 3/4″**	30 / 70	5/16in lever axle

VERSION A

DRILLING (CONTINUED)

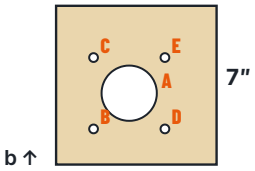
W09 pedestal top



a from the −u edge, b from the −v (right) edge — seen from above

	Hole / bit	a / b (in)	mm	For
A	**1" spade / Forstner** Ø26.2 mm	**4 5/16" / 3 9/16"**	110 / 90	PEX turret tube clearance
B	**1/8"** Ø3.0 mm	**5 3/8" / 4 5/8"**	137 / 117	pilot, #8 screw from the base
C	**1/8"** Ø3.0 mm	**3 1/4" / 4 5/8"**	83 / 117	pilot, #8 screw from the base
D	**1/8"** Ø3.0 mm	**3 1/4" / 2 1/2"**	83 / 63	pilot, #8 screw from the base
E	**1/8"** Ø3.0 mm	**5 3/8" / 2 1/2"**	137 / 63	pilot, #8 screw from the base

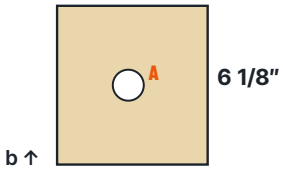
W10 pedestal wall -u (conduit)



a from the −v (right) edge, b from the bottom edge — seen from the +u side

	Hole / bit	a / b (in)	mm	For
A	**2-7/16" hole saw** Ø62.0 mm	**3 1/4" / 3 1/8"**	82 / 80	2in conduit
B	**3/16"** Ø4.6 mm	**1 5/8" / 1 9/16"**	42 / 40	fitting bolt (M3 × 25)
C	**3/16"** Ø4.6 mm	**1 5/8" / 4 3/4"**	42 / 120	fitting bolt (M3 × 25)
D	**3/16"** Ø4.6 mm	**4 13/16" / 1 9/16"**	122 / 40	fitting bolt (M3 × 25)
E	**3/16"** Ø4.6 mm	**4 13/16" / 4 3/4"**	122 / 120	fitting bolt (M3 × 25)

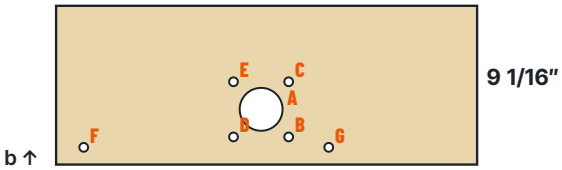
W14 pedestal shelf



a from the −u edge, b from the −v (right) edge — seen from above

	Hole / bit	a / b (in)	mm	For
A	**1-1/4" spade / Forstner** Ø30.4 mm	**2 3/4" / 3 1/16"**	70 / 78	slew-tube-bushing

W18 sandbox wall -u (box side)

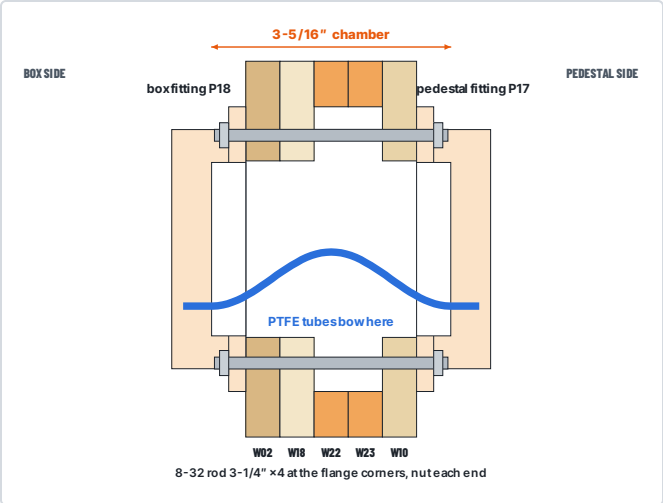


a from the −v (right) edge, b from the bottom edge — seen from the +u side

	Hole / bit	a / b (in)	mm	For
A	**2-7/16" hole saw** Ø62.0 mm	**11 11/16" / 3 1/8"**	297 / 80	2in conduit (seal with silicone)
B	**3/16"** Ø4.6 mm	**13 1/4" / 1 9/16"**	337 / 40	bolt from the control box
C	**3/16"** Ø4.6 mm	**13 1/4" / 4 3/4"**	337 / 120	bolt from the control box
D	**3/16"** Ø4.6 mm	**10 1/8" / 1 9/16"**	257 / 40	bolt from the control box
E	**3/16"** Ø4.6 mm	**10 1/8" / 4 3/4"**	257 / 120	bolt from the control box
F	**3/16"** Ø4.6 mm	**1 9/16" / 1"**	40 / 25	bolt from the control box
G	**3/16"** Ø4.6 mm	**15 9/16" / 1"**	395 / 25	bolt from the control box

VERSION B

MINIMAL VERSION: WHAT CHANGES



Section Version B wall stack on the conduit axis, to scale; each plywood layer 15/32″: W02 box front, W18 sandbox wall, W22 + W23 spacers, W10 pedestal wall. One 8-32 rod clamps everything. The spacers' 3-15/16″ square window holds the rod nuts' far ends clear and gives the PTFE tubes room to bow.

WHY A SPACER, NOT NOTHING DERIVED

The four arm PTFE tubes change length by up to about 7/16″ as the boom and stick move; in version A that spare length coils in the conduit. With the pedestal flat on the sandbox wall there would be only 1-7/8″ between the two fittings, and a tube storing that much there bends to about 5/16″ radius — below the 5/8″ minimum. With the 15/16″ spacer the chamber is 3-5/16″ long and the bow needs about a 3/4″ radius. The chamber is closed on all sides, so no sand gets in.

Estimate from a cosine bow: rise $h = \sqrt{(4 \cdot L \cdot \Delta L)/\pi}$, smallest radius $= L^2 / (2\pi^2 \cdot h)$. Not a CAD check — do the dry fit below.

BUILD NOTES

- Glue W22 to W23, then glue the pair to the outside of the pedestal wall –u (W10), windows lined up with its Ø62 hole.
- Stand the pedestal against the sandbox wall –u, run a bead of silicone round the spacer's outer edge, and clamp with the four rods.
- The arm's slew axis now sits 2-1/4″ closer to the box: 5-1/4″ from the sandbox wall's inside face instead of 7-1/2″.
- **Dry fit:** before closing the box lid, move the boom and stick through their full ranges and check that each stick and bucket tube still slides freely at the turret end.

OTHER STOCK THAT CHANGES DERIVED

Item	A (manual)	B (minimal)
Lever axle (5/16″ rod)	8-1/16″	**7-5/16″**
Slew spool axle (5/16″ rod)	10-13/16″	**7-3/8″**
Lever handles (5/16″ rod) ×3	9-1/16″	**5-11/16″**
2in PVC conduit	5-9/16″	**— (none)**
Fitting bolts	4 × M3 × 35 + 4 × M3 × 25	**4 × 8-32 rod, 3-1/4″, nut each end**
Boom / stick / bucket rope	54-1/2″ / 71-11/16″ / 85-9/16″	**50″ / 67-3/16″ / 81-1/16″**
PTFE stick_hi / stick_lo	25-1/2″ / 25-15/16″	**23-1/4″ / 23-11/16″**
PTFE bucket_hi / bucket_lo	32″ / 32-3/8″	**29-3/4″ / 30-1/8″**
Slew sleeve A / B	13-3/8″ / 16″	**10-3/4″ / 13-3/8″**
Slew rope A / B	32-5/8″ / 35-7/16″	**30-1/16″ / 32-13/16″**
Reprints	—	**slew spool (tube 2-5/16″), lever hubs boom + bucket**

Box rods, handles and reprints from the compact-box spec. Ropes and tubes: the conduit run is 2-1/4″ shorter, so each arm rope (two strands through it) is 4-1/2″ shorter and each tube or sleeve 2-1/4″ shorter; the slew numbers also include the compact box. Cut ropes 3/4″ long and trim when you crimp.

VERSION B

MINIMAL VERSION (NO PVC CONDUIT)

Compact control box 8-7/16″ × 14-1/4″ × 6-1/8″; pedestal 7-3/8″ × 7-1/16″; two spacer frames (W22, W23) in place of the conduit; sandbox unchanged except the bolt holes in W18. Panels generated by `build-box.py` with the compact-box constants and the minimal pedestal.

2294 sq in
plywood in the pieces (15.9 sq ft)

1 × 4×8 ft
sheet to buy (50 % used)

23 pieces
Control box 485 · Pedestal 280 · Sandbox 1430 · Spacer (B) 99 sq in

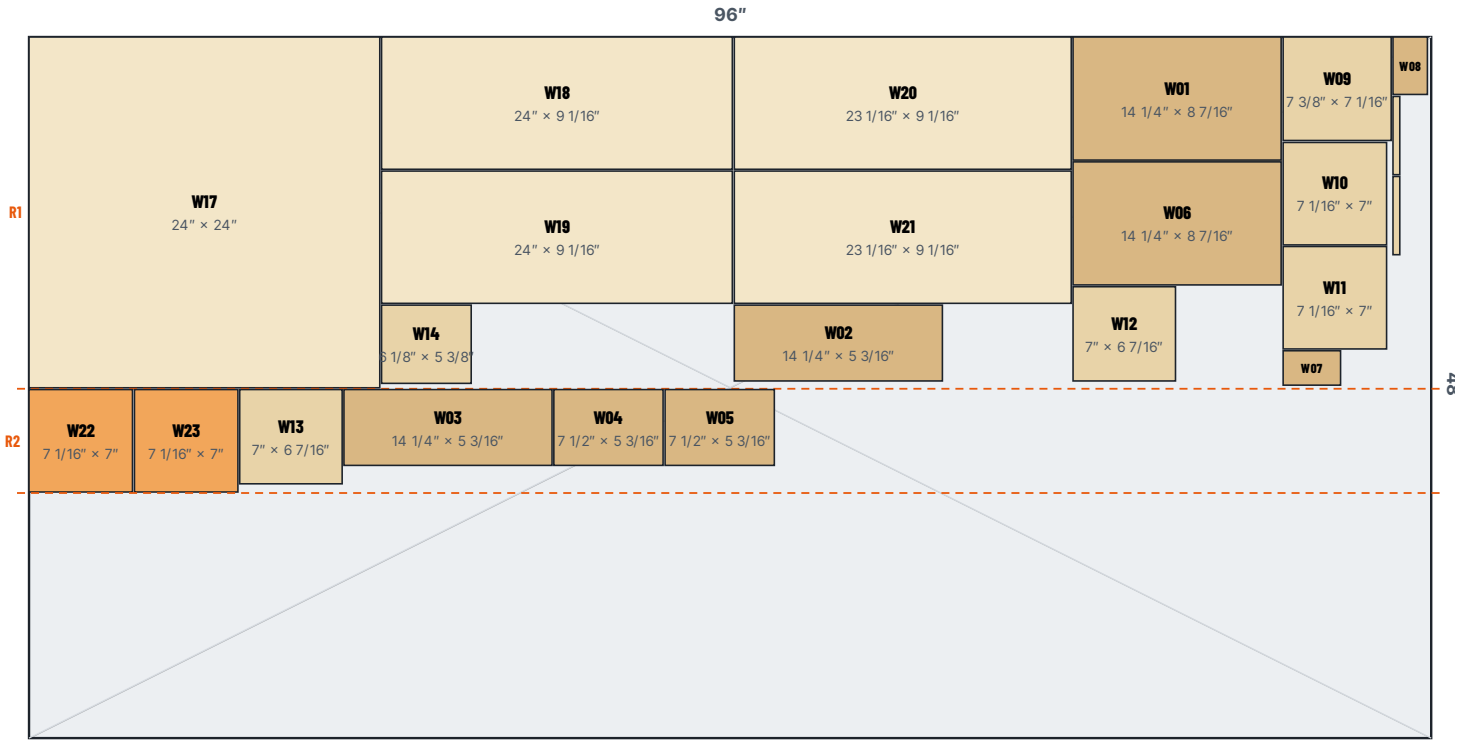
	ID	Piece	Cut size (in)	mm	Holes	sq in
☐	W01	box floor	**14 1/4″ × 8 7/16″**	362 × 214	1	120
☐	W02	box front	**14 1/4″ × 5 3/16″**	362 × 132	7	74
☐	W03	box rear	**14 1/4″ × 5 3/16″**	362 × 132	—	74
☐	W04	box right side	**7 1/2″ × 5 3/16″**	190 × 132	—	39
☐	W05	box left side	**7 1/2″ × 5 3/16″**	190 × 132	—	39
☐	W06	box top (removable)	**14 1/4″ × 8 7/16″**	362 × 214	4	120
☐	W07	axle bearing block left	**3 15/16″ × 2 3/8″**	100 × 60	1	9
☐	W08	axle bearing block right	**3 15/16″ × 2 3/8″**	100 × 60	1	9
☐	W09	pedestal top	**7 3/8″ × 7 1/16″**	188 × 180	5	52
☐	W10	pedestal wall -u (conduit)	**7 1/16″ × 7″**	180 × 178	5	50
☐	W11	pedestal wall +u (removable access)	**7 1/16″ × 7″**	180 × 178	—	50
☐	W12	pedestal wall -v	**7″ × 6 7/16″**	178 × 164	—	45
☐	W13	pedestal wall +v	**7″ × 6 7/16″**	178 × 164	—	45
☐	W14	pedestal shelf	**6 1/8″ × 5 3/8″**	156 × 136	1	33
☐	W15	pedestal shelf cleat -v	**5 3/8″ × 1/2″**	136 × 12	—	3
☐	W16	pedestal shelf cleat +v	**5 3/8″ × 1/2″**	136 × 12	—	3
☐	W17	sandbox floor	**24″ × 24″**	610 × 610	—	577
☐	W18	sandbox wall -u (box side)	**24″ × 9 1/16″**	610 × 230	7	217
☐	W19	sandbox wall +u	**24″ × 9 1/16″**	610 × 230	—	217
☐	W20	sandbox wall -v	**23 1/16″ × 9 1/16″**	586 × 230	—	209
☐	W21	sandbox wall +v	**23 1/16″ × 9 1/16″**	586 × 230	—	209
☐	W22	spacer frame 1	**7 1/16″ × 7″**	180 × 178	1	50
☐	W23	spacer frame 2	**7 1/16″ × 7″**	180 × 178	1	50

VERSION B

SHEET LAYOUT AND CUT ORDER

Rip the strips first (orange dashed lines, R1, R2 …), then crosscut each strip. Every piece is drawn to scale with 1/8″ kerf between pieces; grey is offcut. Mark each piece with its ID as soon as it is cut.

SHEET 1 OF 1 — 4 × 8 FT, 1/2″ PLYWOOD



Thin strips not labelled on the drawing: W15, W16 — see the cut order.

R1: rip a strip **24″** wide; crosscut: 24″ → **W17**; 24″ → **W18** + **W19** + **W14** (rip that piece into 9 1/16″ / 9 1/16″ / 5 3/8″); 23 1/16″ → **W20** + **W21** + **W02** (rip that piece into 9 1/16″ / 9 1/16″ / 5 3/16″); 14 1/4″ → **W01** + **W06** + **W12** (rip that piece into 8 7/16″ / 8 7/16″ / 6 7/16″); 7 3/8″ → **W09** + **W10** + **W11** + **W07** (rip that piece into 7 1/16″ / 7″ / 7″ / 2 3/8″); 2 3/8″ → **W08** + **W15** + **W16** (rip that piece into 3 15/16″ / 5 3/8″ / 5 3/8″).

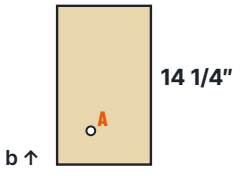
R2: rip a strip **7″** wide; crosscut: 7 1/16″ → **W22**; 7 1/16″ → **W23**; 7″ → **W13**; 14 1/4″ → **W03**; 7 1/2″ → **W04**; 7 1/2″ → **W05**.

VERSION B

DRILLING

Measure **a** along the bottom edge and **b** up the side, from the lower-left corner as drawn. Holes: the bold size is the drill bit to use. Clamp a scrap under the panel when drilling the big holes (2-7/16″ hole saw).

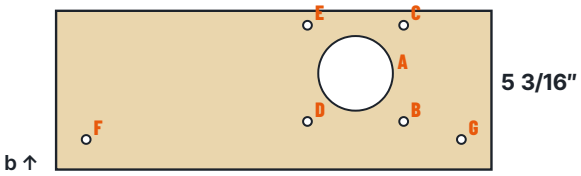
W01 box floor



a from the −u edge, b from the −v (right) edge — seen from above

	Hole / bit	a / b (in)	mm	For
A	**11/32″** Ø8.5 mm	**3 1/16″ / 3 1/16″**	77 / 77	slew spool axle (nut + washer in a Ø22 × 7 counterbore from below)

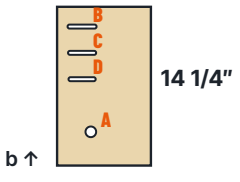
W02 box front



a from the −v (right) edge, b from the bottom edge — seen from the +u side

	Hole / bit	a / b (in)	mm	For
A	**2-7/16″ hole saw** Ø62.0 mm	**9 13/16″ / 3 1/8″**	249 / 80	2in conduit
B	**3/16″** Ø4.6 mm	**11 3/8″ / 1 9/16″**	289 / 40	8-32 through-rod
C	**3/16″** Ø4.6 mm	**11 3/8″ / 4 3/4″**	289 / 120	8-32 through-rod
D	**3/16″** Ø4.6 mm	**8 1/4″ / 1 9/16″**	209 / 40	8-32 through-rod
E	**3/16″** Ø4.6 mm	**8 1/4″ / 4 3/4″**	209 / 120	8-32 through-rod
F	**3/16″** Ø4.6 mm	**1″ / 1″**	25 / 25	box-to-sandbox bolt (M3 × 30)
G	**3/16″** Ø4.6 mm	**13 1/4″ / 1″**	337 / 25	box-to-sandbox bolt (M3 × 30)

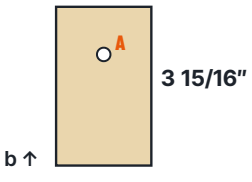
W06 box top (removable)



a from the −u edge, b from the −v (right) edge — seen from above

	Hole / bit	a / b (in)	mm	For
A	**1″ spade / Forstner** Ø24.4 mm	**3 1/16″ / 3 1/16″**	77 / 77	spool tube bearing
B	**slot 2 5/8″**: 23/64″ holes at a 1 1/8″ & 3 3/8″, saw between 66.1 mm	**2 1/4″ / 12 1/2″**	57 / 317	boom lever slot = hard stop (42 deg swing)
C	**slot 2 1/2″**: 23/64″ holes at a 1 3/16″ & 3 5/16″, saw between 64.1 mm	**2 1/4″ / 10 1/8″**	57 / 257	stick lever slot = hard stop (41 deg swing)
D	**slot 2 7/16″**: 23/64″ holes at a 1 3/16″ & 3 5/16″, saw between 61.8 mm	**2 1/4″ / 7 3/4″**	57 / 197	bucket lever slot = hard stop (40 deg swing)

W07 + W08 axle bearing blocks (×2, identical)



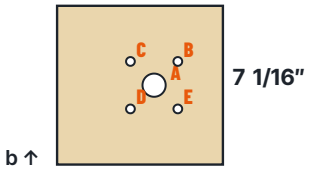
a from the −u edge, b from the bottom edge — seen from the −v side

	Hole / bit	a / b (in)	mm	For
A	**11/32″** Ø8.5 mm	**1 3/16″ / 2 3/4″**	30 / 70	5/16in lever axle

VERSION B

DRILLING (CONTINUED)

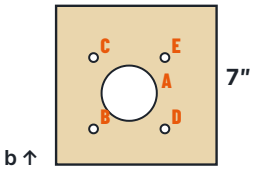
W09 pedestal top



a from the −u edge, b from the −v (right) edge — seen from above

	Hole / bit	a / b (in)	mm	For
A	**1″ spade / Forstner** Ø26.2 mm	**4 5/16″ / 3 9/16″**	110 / 90	PEX turret tube clearance
B	**1/8″** Ø3.0 mm	**5 3/8″ / 4 5/8″**	137 / 117	pilot, #8 screw from the base
C	**1/8″** Ø3.0 mm	**3 1/4″ / 4 5/8″**	83 / 117	pilot, #8 screw from the base
D	**1/8″** Ø3.0 mm	**3 1/4″ / 2 1/2″**	83 / 63	pilot, #8 screw from the base
E	**1/8″** Ø3.0 mm	**5 3/8″ / 2 1/2″**	137 / 63	pilot, #8 screw from the base

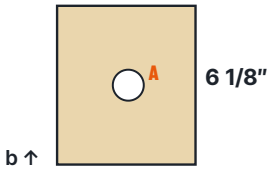
W10 pedestal wall -u (conduit)



a from the −v (right) edge, b from the bottom edge — seen from the +u side

	Hole / bit	a / b (in)	mm	For
A	**2-7/16″ hole saw** Ø62.0 mm	**3 1/4″ / 3 1/8″**	82 / 80	2in conduit
B	**3/16″** Ø4.6 mm	**1 5/8″ / 1 9/16″**	42 / 40	8-32 through-rod
C	**3/16″** Ø4.6 mm	**1 5/8″ / 4 3/4″**	42 / 120	8-32 through-rod
D	**3/16″** Ø4.6 mm	**4 13/16″ / 1 9/16″**	122 / 40	8-32 through-rod
E	**3/16″** Ø4.6 mm	**4 13/16″ / 4 3/4″**	122 / 120	8-32 through-rod

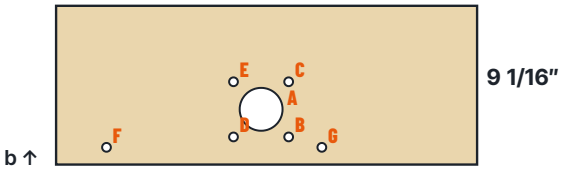
W14 pedestal shelf



a from the −u edge, b from the −v (right) edge — seen from above

	Hole / bit	a / b (in)	mm	For
A	**1-1/4″ spade / Forstner** Ø30.4 mm	**2 3/4″ / 3 1/16″**	70 / 78	slew-tube-bushing

W18 sandbox wall -u (box side)



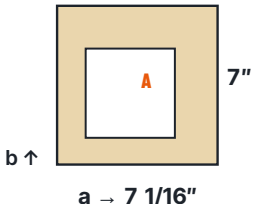
a from the −v (right) edge, b from the bottom edge — seen from the +u side

	Hole / bit	a / b (in)	mm	For
A	**2-7/16″ hole saw** Ø62.0 mm	**11 11/16″ / 3 1/8″**	297 / 80	2in conduit (seal with silicone)
B	**3/16″** Ø4.6 mm	**13 1/4″ / 1 9/16″**	337 / 40	8-32 through-rod
C	**3/16″** Ø4.6 mm	**13 1/4″ / 4 3/4″**	337 / 120	8-32 through-rod
D	**3/16″** Ø4.6 mm	**10 1/8″ / 1 9/16″**	257 / 40	8-32 through-rod
E	**3/16″** Ø4.6 mm	**10 1/8″ / 4 3/4″**	257 / 120	8-32 through-rod
F	**3/16″** Ø4.6 mm	**2 7/8″ / 1″**	73 / 25	bolt from the control box
G	**3/16″** Ø4.6 mm	**15 3/16″ / 1″**	385 / 25	bolt from the control box

VERSION B

DRILLING (CONTINUED)

W22 + W23 spacer frames (×2, identical)



a from the −v (right) edge, b from the bottom edge — seen from the +u side

	Hole / bit	a / b (in)	mm	For
A	**window 3 15/16″ square**: corners at a 1 1/4″ – 5 3/16″, b 1 3/16″ – 5 1/8″ 100 mm	**3 1/4″ / 3 1/8″**	82 / 80	window (drill 3/8″ in each corner, then jigsaw)